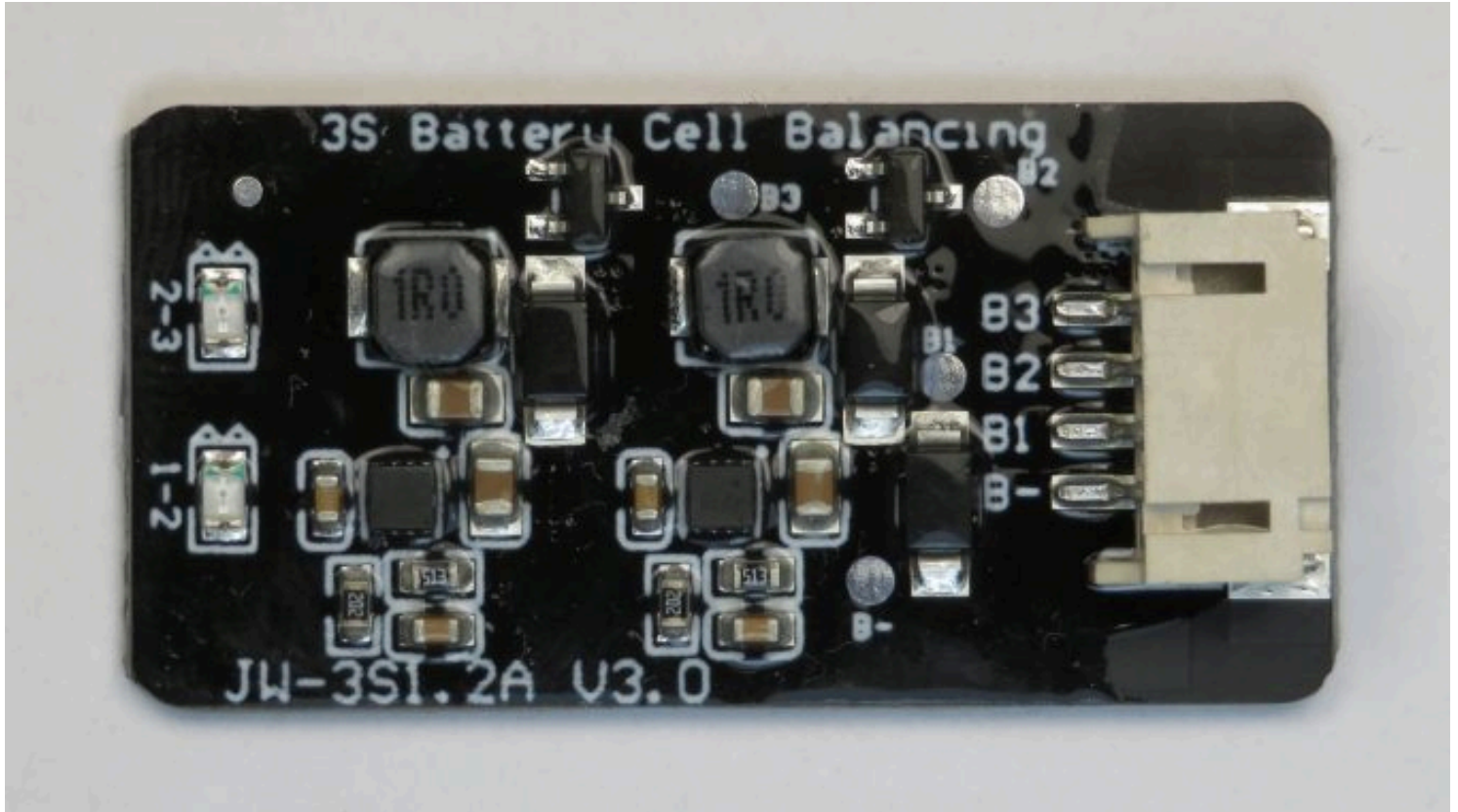


Review: Li-ion LiPo LiFePO4 Lithium Battery Active Equalizer Balancer Energy Transfer Board



Cheap Lithium Battery “Active Balancer” boards have been turning up on Aliexpress and other eCommerce sites.

But there has been some scepticism if they are genuinely active balancing and quite a few conflicting forum posts. Some have even claimed the ‘1R0’ inductors are actually 1 ohm resistors.

The more traditional passive balancer boards will simply dump any excessive charge into a resistor, wasting this energy. For example, if the maximum voltage threshold was set at 4.2V, once the cell’s terminal voltage exceeded 4.2V, any excess charge would be dumped into a resistor and dissipated as heat until the terminal voltage fell below 4.2V.

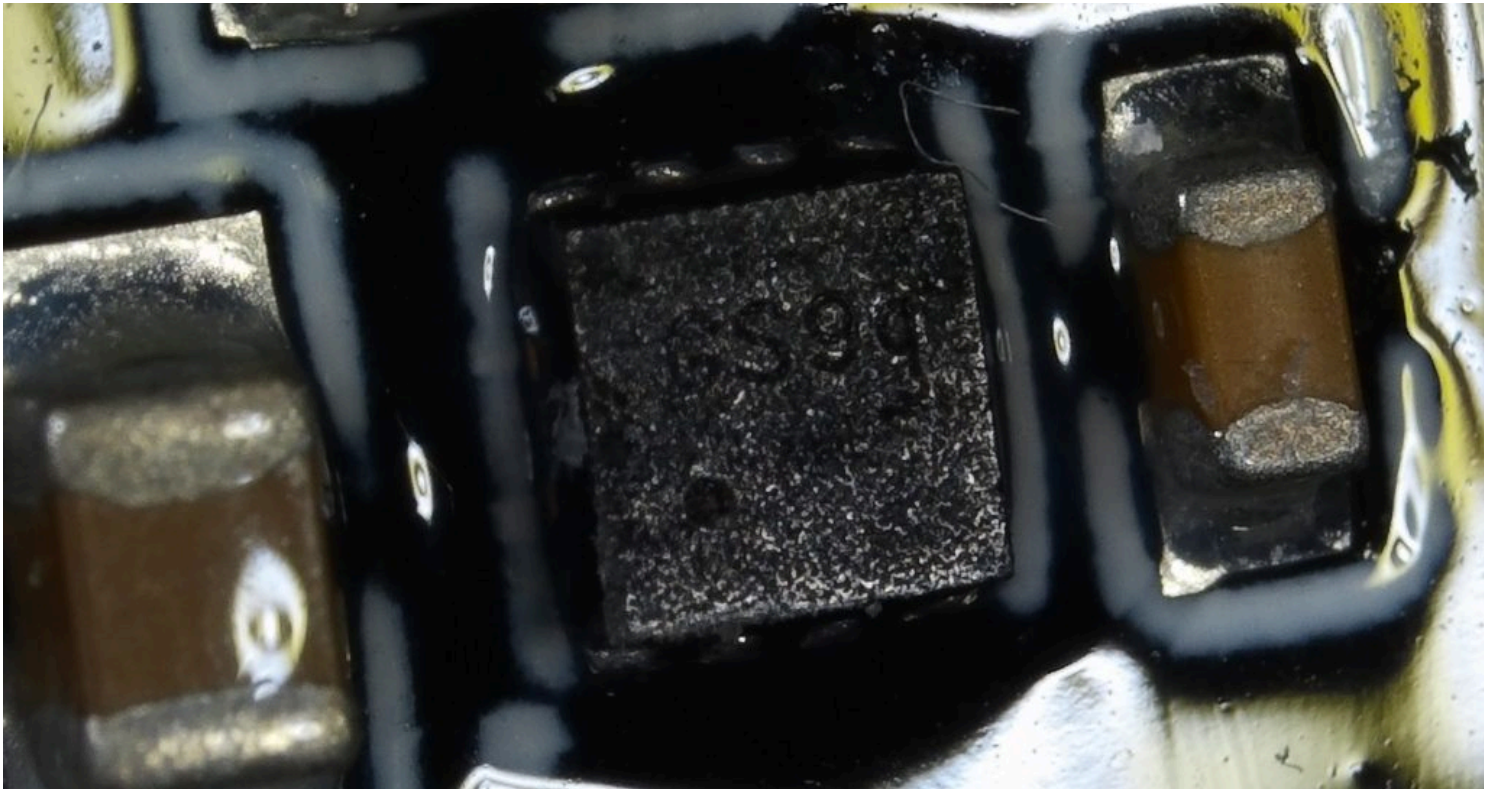
Active balancer boards are designed to divert excess energy from the cells with the highest voltage into cells that still require charge.

To find out how these cheap boards work, I jumped in and paid \$8.26 USD for a **3S version from the Aliexpress Dykbhuang Store**.

This board has a PCB part number of JW-3S1.2A V3.0. By decoding the part number, the designer appears to have the initials JW. This is a 3S version 3.0 PCB with a maximum balancing capability of 1.2A.

ETA3000 Inductive Cell Balancer IC.

The easiest way to understand how they work would be to ascertain the actual integrated circuit (I.C.) used in the design and have a study of the datasheet. The board was conformal coated, but scratching off the coating reveal a marking of GS9q.



The active integrated circuit had a component mark that looked like GS9q

After quite a bit of searching, I would come to the conclusion that it is a ETA3000 Inductive Cell Balancer IC from Shanghai manufacturer [ETA Solutions](#).

The part doesn't seem to be listed on their website, but the datasheet can be found [here](#), thanks to Google.

This device comes in two packages, a SOT23-6 or DFN.

The DFN version, part number ETA3000D2I has a marking GSYW where it appears the Y is the Year and W is the week. This would suggest my parts were made in 2019, but I'm uncertain about the week datecode.

Schematic

The ETA solutions datasheet provides the typical circuit:

TYPICAL APPLICATION

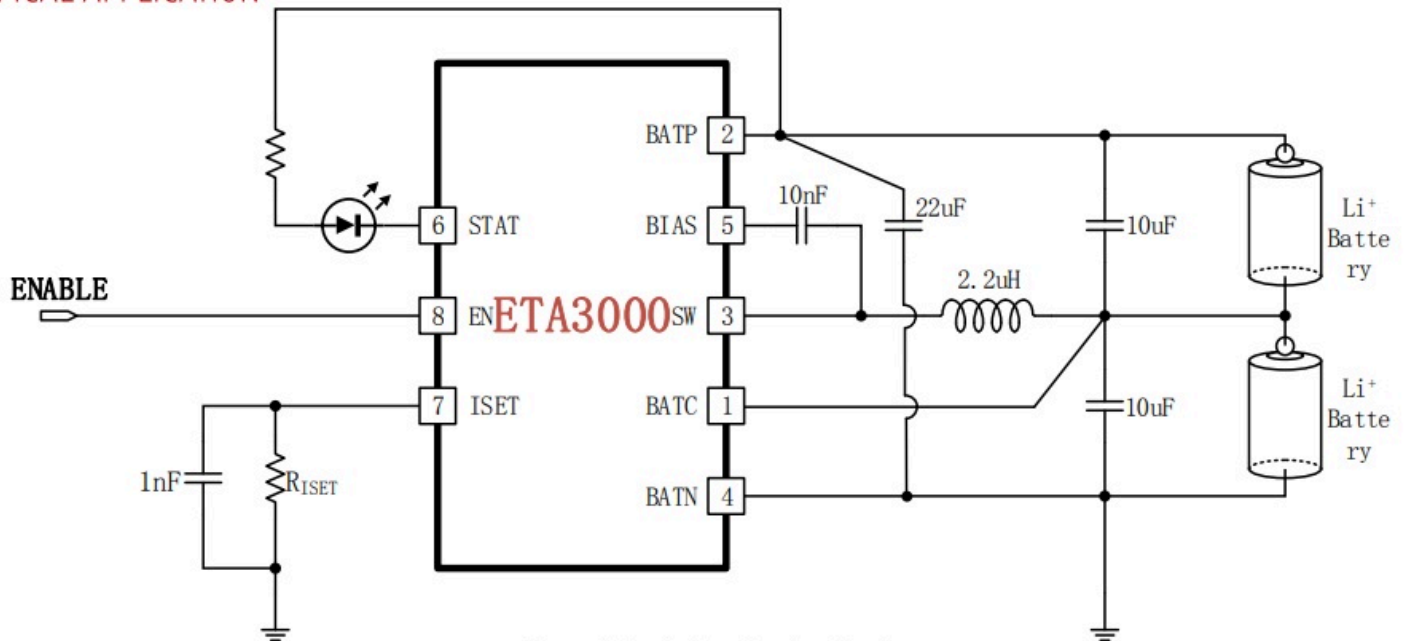


Figure 1: Typical Application Circuit

Typical schematic extracted from ETA3000 datasheet.

According to the datasheet, the “ETA3000 is a battery cell balancer with lossless inductive architecture based on ETA’s proprietary technology.”

The device will shuffle current between the two cells until the cells are balanced. To do this, it uses an inductor switching at 1MHz. (The **1R0 marking** means the SMD inductor is 1uH)

To begin the “shuffle”, the cells must exceed the unbalanced detection threshold of 100mV and be within the Under-Voltage Lockout (UVLO) and Over-Voltage Protection (OVP) thresholds. This is between 3.75 and 5V.

The balancing will terminate when the cells are balanced. Balance accuracy is specified as +/-30mV.

When not balancing the device goes to sleep. Sleep current is specified as 2uA.

Layout

The layout is consistent with the recommendation from ETA solutions. The 10nF bootstrap cap has been placed on the left hand side of the ETA3000.

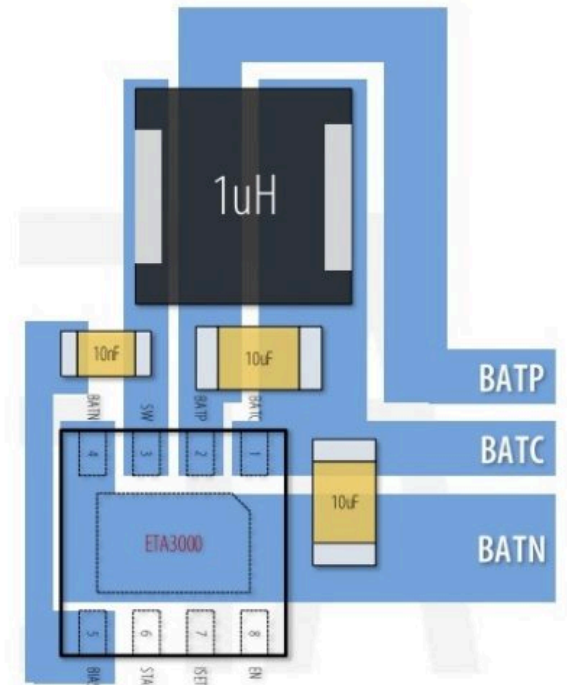
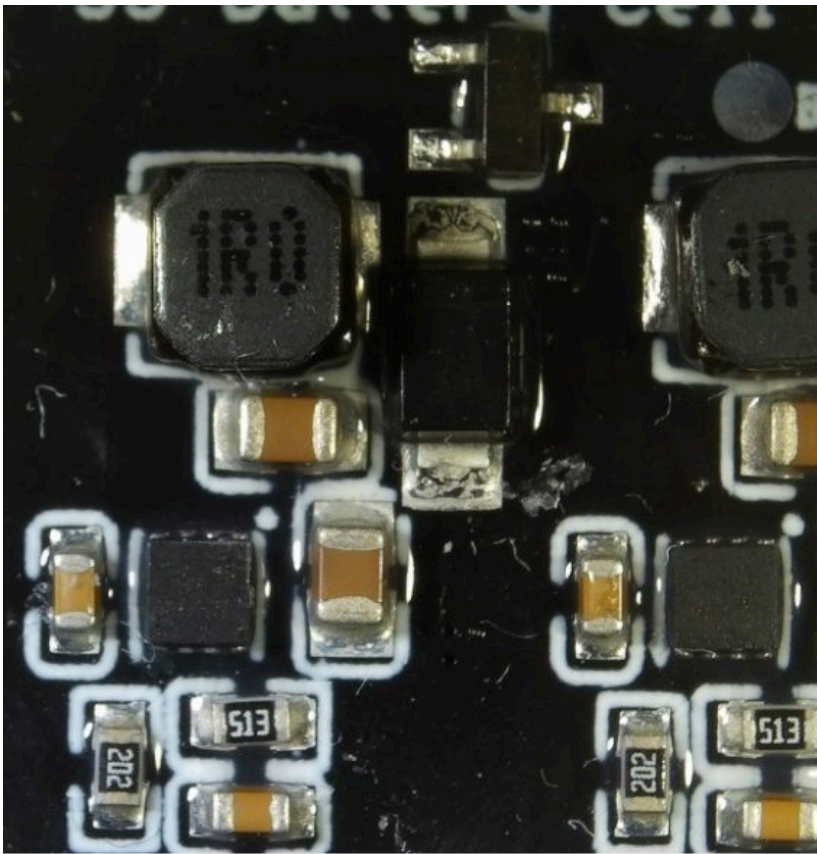


Figure 8: PCB Recommendation, DFN2X2-8L

The 51k resistor (513) and 1nF capacitor below forms the programmable current setting. The 51k sets the maximum current to just short of 1A. (My Aliexpress listing said 1.2A in the title, and 1.5A in the description/features)

One would assume the 2k resistor close to the STAT pin is the LED current limiting resistor.

Multicell Balancing

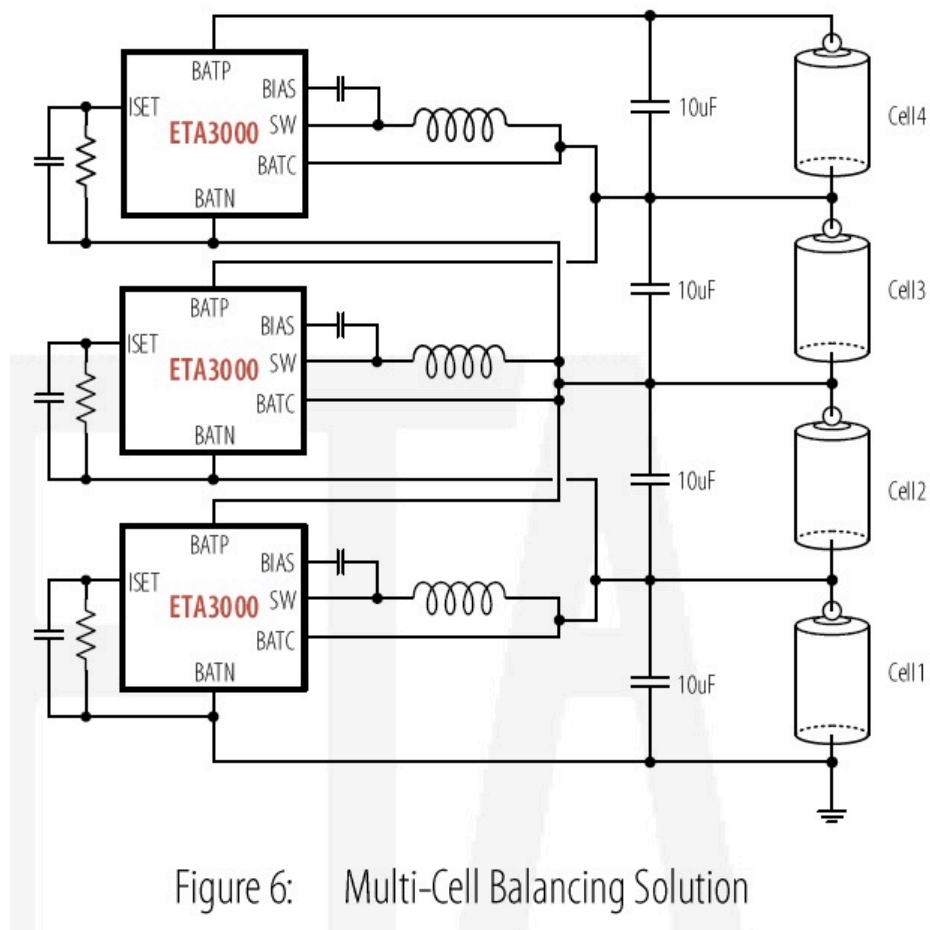


Figure 6: Multi-Cell Balancing Solution

Multi-Cell Balancing Functional Schematic from ETA Solutions Datasheet.

The topology allows overlapping ETA3000 to be connected together to balance multiple cell battery packs. Merchants on Aliexpress are currently selling boards for battery packs from 3S to 16S.

Maximum transfer efficiency is specified at *up to* 92% per device. Charge has to be passed through sequential cells, but one could argue this is still more efficient than wasting all the energy in heat via a resistor.

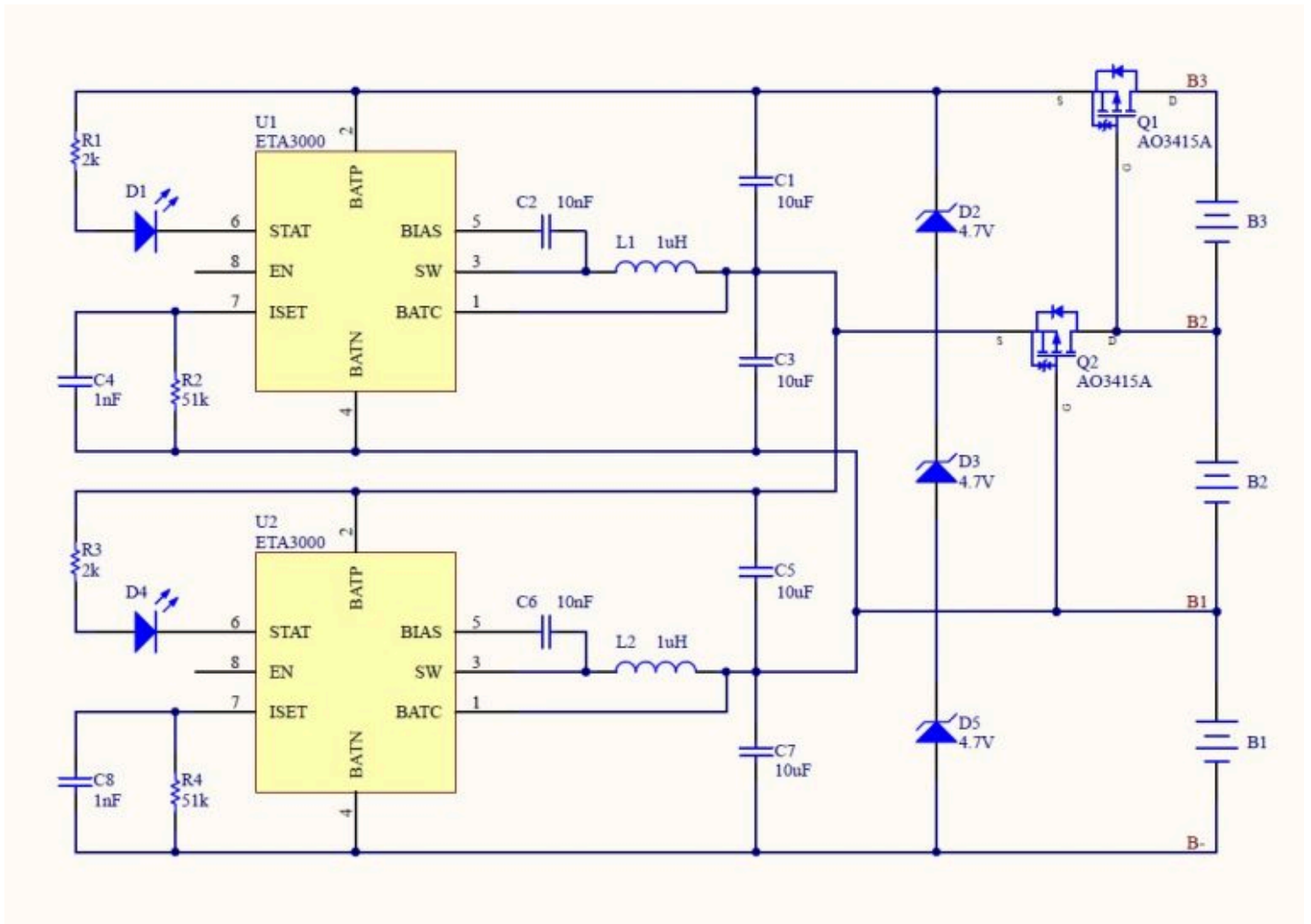
It should be noted, the balancing accuracy of $\pm 30\text{mV}$ is between the two cells connected to a single ETA3000 and not over the entire board.

In a worse case scenario, it is possible for these errors to accumulate between cells of the pack. For example, the first cell could be 4.10V, the next cell 4.07V, then 4.04V etc.

However battery packs should be made of identical and matched cells that rarely need balancing, assuming they are not abused – i.e. depleted beyond their low cutoff voltages. Hence, only one or two cells in a pack should ever need balancing.

Extra Protection Circuitry

In addition to the typical circuit, extra protection has been added to this design. (*Thanks JW*)



The large diodes on the PCB appears to be a 4.7V 1W Zener and is designed to provide some protection for when a cell voltage exceeds the breakdown/knee voltage. I say “some” protection assuming cell taps are fused and not low impedance connections straight into the battery.

The SOT-23 parts also forms as added protection. They are AO3415 20V P-Channel MOSFET (marking AF9TF J) and provides reverse voltage protection on cell taps B2 & B3 (*on my 3S V3.0 board*).

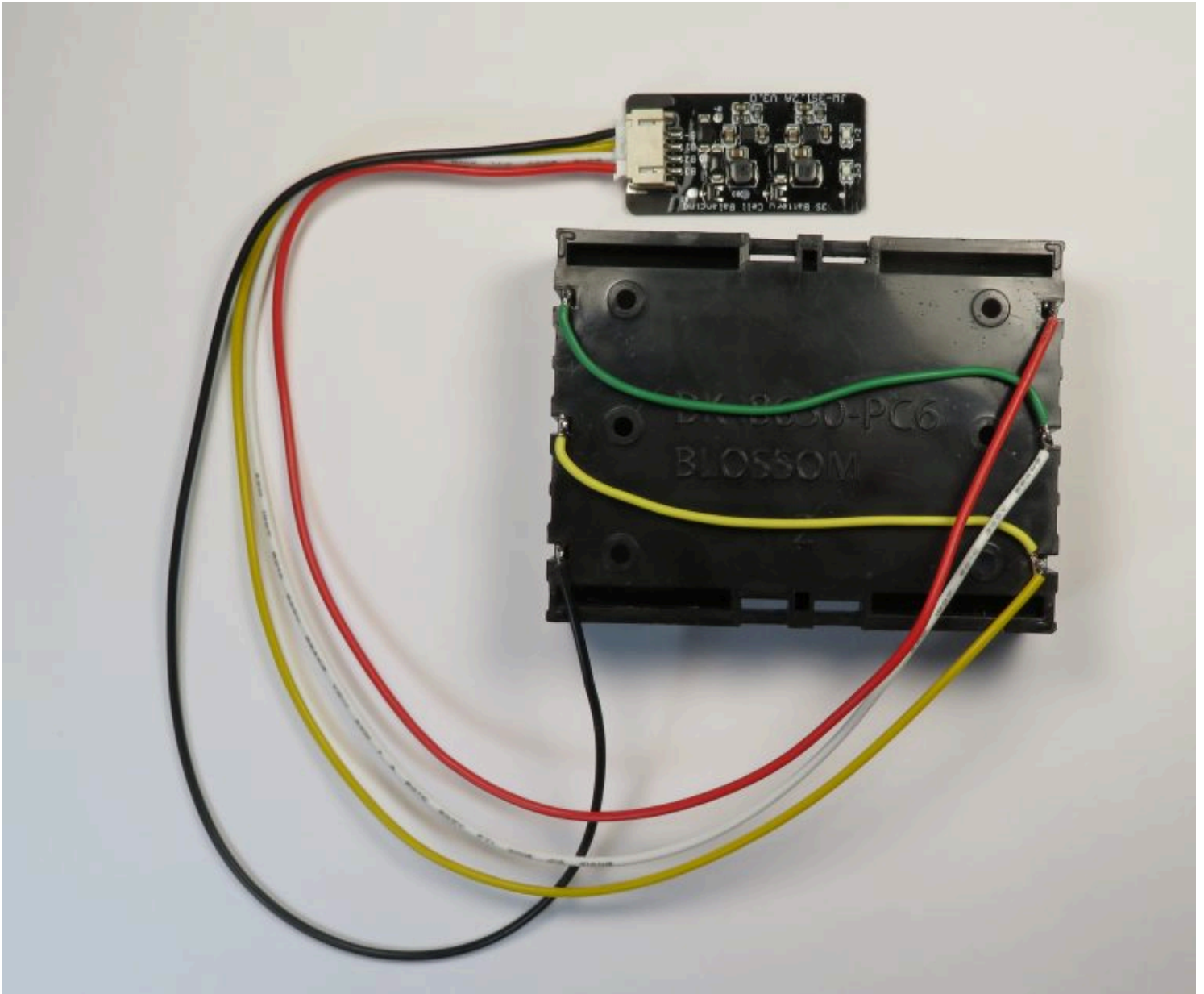
I have yet to work out why cell tap B1 is not reverse voltage protected. While this input does not power either ETA3000 (the chip is powered via BATP), the datasheet does indicate the absolute maximum rating of BATC to BATN is -0.3V to 6V, suggesting the BATC pin is not tolerant to reverse voltages.

Testing

The desktop study would suggest the board is the real deal. However to date, I have only done very preliminary testing.

The Lithium Battery Active Equalizer Balancer Energy Transfer Board comes complete with cable to bare-ends. Please note, this connector is not a JST-HX balance connector that you would commonly find on hobby

packs.



I have wired it to a 18650 battery holder so I can switch in and out batteries of various states of charge.

So far the board has successfully balanced the batteries to within 30mV between adjacent cells.

More comprehensive testing is to come